ITDS 124 Mathematics of Computing

Chapter 1. Fundamentals

1.3 Sequences

Created by : Rawesak Tanawongsuwan
rawesak.tan@mahidol.ac.th

Sequence

- A sequence is a list of objects arranged in a definite order.
- Differences between set and sequence
  - A set has no order and no duplicated elements
  - A sequence has a specific order and elements may be duplicated
- A finite sequence is a sequence that stops after n steps, $n \in \mathbb{N}$.
  - $1,0,0,1,0,1,1,1,1$
- An infinite sequence is a sequence that goes on forever.
  - $3,8,13,18,23, \ldots$

Sequence formula

- **Recursive** : a formula that refers to previous terms to define the next term
  - Note that every recursive term formula must include a starting place
  - The recursive formula $c_1 = 5,\ c_n = 2c_{n-1},\ 2 \le n \le 6$ defines the finite sequence $5,10,20,40,80,160.$
  - The recursive formula $d_1 = 3,\ d_n = d_{n-1} + 4$ defines the infinite sequence $3,7,11,15,19,23, \ldots$
- **Explicit** : a formula that tells us exactly what value any particular term has
  - The explicit formula $s_n = (-4)^n,\ 1 \le n,$ defines the infinite sequence $-4, 16, -64, 256, \ldots$
  - The explicit formula $t_n = 92 - 5n,\ 1 \le n \le 5$ defines the finite sequence $87,82,77,72,67$

Strings (สายอักขระ)

- Sequences of letters or other symbols, written without the commas.

- Examples :
  - An ordinary English word such as "*study*" can be viewed as the finite sequence s, t, u, d, y
  - An infinite string such as $abababab$ … can be viewed as the infinite sequence $a, b, a, b, a, b, a, b, \ldots$
  - The sentence "*I am a student*" can be viewed as a finite sequence of words: $I, am, a, student$ ← These elements of the sequence are words of varying length.

The set corresponding to a sequence

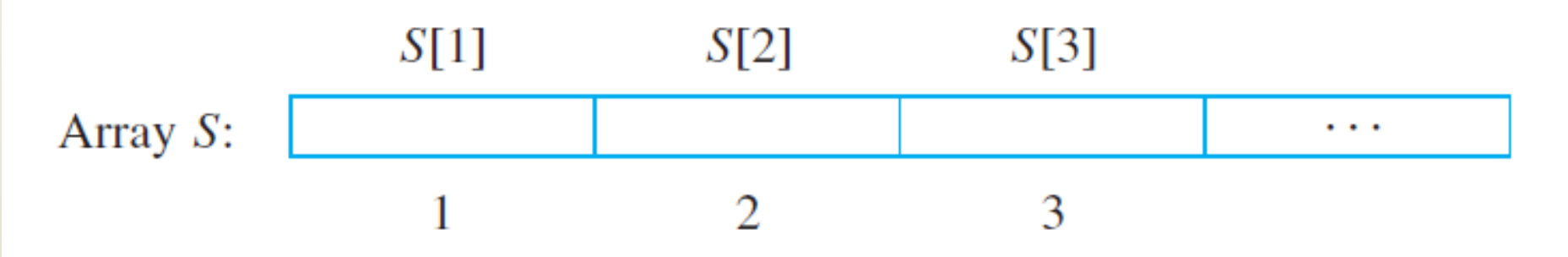
- The set of all distinct elements in the sequence.

- The set corresponding to the sequence $1,3,5,7,9,\ldots$ is $\{1,3,5,7,9,\ldots\}$.
- The set corresponding to the sequence $abcabcabcabc\ldots$ is $\{a,b,c\}$.

Sequence as an Array

- A sequence is sometimes called a linear array or list.

- A sequence $S : s_1, s_2, s_3, \ldots, s_n$
- An array $S : S[1], S[2], S[3], \ldots S[n]$ ← a sequence of boxes



Characteristic functions

- If A is a subset of a universal set U, the characteristic function f_A of A is defined for each $x \in U$ as follows :

$$f_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

- Ex. Let $U = \{1,2,3,4,5,6\}$, $A = \{1,2\}$, $B = \{2,4,6\}$, $C = \{4,5,6\}$.
  - $f_A(1) = 1$, $f_A(2) = 1$, $f_A(6) = 0$, $f_A = \{1,1,0,0,0,0\}$
  - $f_B(1) = 0$, $f_B(2) = 1$, $f_B(6) = 1$, $f_B = \{0,1,0,1,0,1\}$
  - $f_C(1) = 0$, $f_C(2) = 0$, $f_C(6) = 1$, $f_C = \{0,0,0,1,1,1\}$

Properties of characteristic functions of subsets

- $f_{A\cap B} = f_A f_B$

  that is $f_{A\cap B}(x) = f_A(x) f_B(x)$ for all x.

- $f_{A\cup B} = f_A + f_B - f_A f_B$

  that is $f_{A\cup B}(x) = f_A(x) + f_B(x) - f_A(x) f_B(x)$ for all x.

- $f_{A\oplus B} = f_A + f_B - 2 f_A f_B$

  that is $f_{A\oplus B}(x) = f_A(x) + f_B(x) - 2 f_A(x) f_B(x)$ for all x.

Countable and Uncountable

- A set is countable if it corresponds to some sequence.
  - Members of set can be arranged in a list
  - Elements have positions
- All finite sets are countable.
- Many infinite sets are countable. However, there are also infinite sets that are uncountable.
  - For example, a set of real numbers is uncountable, i.e.what comes after 1.23534 ?

Strings and Regular Expressions

- Given a set A, the set A^* consists of all finite sequences of elements of A
- Example:
  - $A = \{a, b, c, \dots, z\} = \text{the set of English alphabet}$
  - A^* = words (the finite sequences, words, in A are not written with commas)
  - A^* contains all possible words, even those that are unpronounceable or make no sense such as “prsartkc”, “tttxxxyyyz”
- The empty sequence or empty string (contain no symbol) is represented with Λ (lambda), λ (lambda), or ε (epsilon).

Catenation (or Concatenation)

- Two strings may be joined into a single string
- If $w_1 = s_1 s_2 s_3 \ldots s_n$ and $w_2 = t_1 t_2\ t_3 \ldots\ t_k$ are elements of A^* for some set A, we define the catenation of w_1 and w_2 as the sequence $s_1 s_2 s_3 \ldots s_n t_1 t_2\ t_3 \ldots\ t_k$.
- The catenation of w_1 and w_2 is written as $w_1 \cdot w_2$ or $w_1 w_2$, and is another element of A^*.
- Note that if w belongs to A^*, then $\mathrm{w} \cdot \Lambda = \Lambda \cdot w = w$.

Example

- Let $A = \{John, Sam, Jane, swims, runs, well, quickly, slowly\}$.
- A sequence $"Jane\ swims\ quickly" \in A^*$
  - Note that we use space to separate the elements in each sequence
- A sequence $"Sam\ runs\ well" \in A^*$
- A sequence $"Well\ swims\ Jane\ slowly\ John" \in A^*$

Regular expression (การกำหนดรูปแบบอักขระ)

- A regular expression (RE) over A is a string constructed from the elements of A and the symbols $(,\ ),\ \vee,\ *,\ \Lambda,$ according to the following definition

RE1. The symbol Λ is a regular expression.

RE2. If $x \in A$, the symbol x is a regular expression.

RE3. If α and β are regular expressions, then the expression $\alpha\beta$ is regular.

RE4. If α and β are regular expressions, then the expression $(\alpha \vee \beta)$ is regular.

RE5. If α is a regular expressions, then the expression $(\alpha)^*$ is regular.

Note that RE1 and RE2 provide initial regular expressions. RE3, RE4, RE5 are used repetitively to define larger sets of regular expressions, so they are recursive formula

Example

Let $A = \{0,1\}$. Show that the following expressions are all regular expressions over A.

- $0^*(0 \vee 1)^*$

  By RE2, 0 and 1 are regular expressions

  By RE4, $(0 \vee 1)$ is regular expression

  By RE5, 0^* and $(0 \vee 1)^*$ are regular expressions

  By RE3, $0^*(0 \vee 1)^*$ is regular expression

- Try $00^*(0 \vee 1)^*1$ and $(01)^*(01 \vee 1^*)$

Regular sets

- Each regular expression over A, there is a corresponding subset of A^*.
- Those sets are called regular subsets of A^* or just regular sets.
- To compute the regular set of the given regular expression, use these rules

1. The expression Λ corresponds to the set $\{\Lambda\}$, where Λ is the empty string in A^*.
2. If $x \in A$, then the regular expression x corresponds to the set $\{x\}$.
3. If α and β are regular expressions corresponding to the subsets M and N of A^*, then $\alpha\beta$ corresponds to $M \cdot N = \{s \cdot t \mid s \in M \text{ and } t \in N\}$. Then $M \cdot N$ is the set of all catenations of strings in M with strings in N.
4. If the regular expressions α and β correspond to the subsets M and N of A^*, then $(\alpha \vee \beta)$ corresponds to $M \cup N$.
5. If the regular expression α corresponds to the subset M of A^*, then $(\alpha)^*$ corresponds to the set M^*.

Example 1

- Let $A = \{a, b, c\}$.
  - The regular expression a^* corresponds to the set of all finite sequences of a's such as $aaa, aaaaaaa$, and so on.
  - The regular expression $a(b \vee c)$ corresponds to the set $\{ab, ac\}$.
  - The regular expression $ab(bc)^*$ corresponds to the set of all strings that begin with ab, and then repeat with bc n times, where $n \geq 0$. Regular set is $\{ab, abbc, abbcbc, \ldots\}$

Example 2

- Let $A = \{0,1\}$. Find regular sets corresponding to the following regular expressions.
- $0^*(0 \vee 1)^*$
  - Regular set $R =$
- $00^*(0 \vee 1)^*1$
  - Regular set $S =$
- $(01)^*(01 \vee 1^*)$
  - Regular set T =